

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

Which fragments could grow into a piece of music with the ones already in a musician’s open project? This target has no direct labels at cold start, and audio similarity is only one observable signal for it. Pocket Producer, a DAW prototype evaluated offline, treats this missing-label problem as a ranking and an interaction-design challenge. It fingerprints the open session and ranks candidates with a rules default and a selectable fusion of five interpretable signals trained on pack co-membership as cold-start weak supervision. On the Freesound Loop Dataset, the learned fusion’s apparent advantage varies with label construction, negative regime, and representation/training configuration. Because pack co-membership is only a proxy for this target, and conclusions vary with how the proxy is constructed and evaluated, retrieval is surfaced as *proposals* the musician can inspect, preview, insert, and undo, rather than automated decisions.

1. INTRODUCTION

Pocket Producer addresses **session compatibility**: whether a library fragment is promising material for the same piece of music as the fragments already in the open project (the *session*). Loop compatibility and sample retrieval [1, 2], stem coherence and retrieval [3, 4], DAW-integrated and context-conditioned generation [5, 6], and human-in-the-loop co-creation [7] address parts of this problem. Unlike pairwise compatibility [1], this target is conditioned on the surrounding session material, as in context-based retrieval [2, 4]. Audio similarity is one observable signal for this target, not its definition. With no direct measurement at cold start, we use pack co-membership as weak supervision. Prior work has validated related proxies against human judgment [1, 2]. Here, we test how ranking conclusions change with the construction and evaluation of the proxy.

We contribute: (1) a framing of DAW fragment retrieval as ranking for unobserved session compatibility under cold-start weak supervision; (2) a sensitivity analysis across three axes (label construction, negative regime,

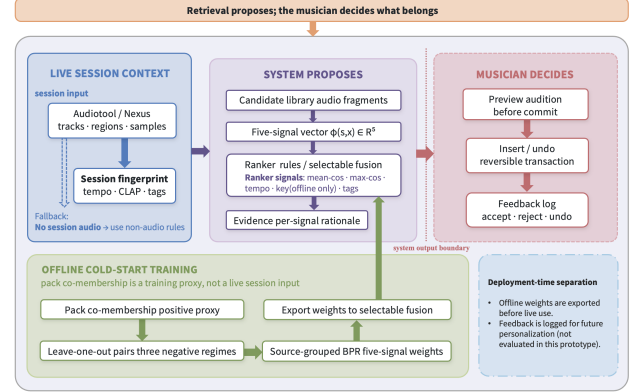


Figure 1. Pocket Producer architecture; pack co-membership is a training proxy, not a live session signal. and representation/training configuration), showing how learned fusion’s relative performance changes; and (3) an offline-evaluated Audiotoool prototype that presents retrieved fragments as inspectable, reversible proposals, with interaction logging as a path toward direct supervision.

2. POCKET PRODUCER

2.1 Session Fingerprinting and Ranking

Pocket Producer (Figure 1) reads the Audiotoool session via the Nexus SDK (offline fixtures; live-account capture pending): the project tempo, the fragments’ tags, and up to eight unique region-referenced samples embedded with CLAP [8] together form the session fingerprint.

Offline, five interpretable signals score loop-only FSLD [9] candidates (deployment: Section 2.3) in the context of the open session: (i) mean-pooled and (ii) max-pooled CLAP cosine similarity to session samples, (iii) tempo proximity, (iv) key match to the session fragments’ FSLD key metadata, and (v) tag Jaccard overlap with the session fragments’ tags. Writing $\phi(s, x) \in \mathbb{R}^5$ for these signals given session s and candidate x , the fusion ranks by $w^\top \phi(s, x)$, with w trained by BPR [10] on pack co-membership weak labels, negatives drawn from a mixture of regime-specific distributions (Section 3.1). A hand-tuned *rules* ranker over non-audio signals (tempo, key, tags, stated intent, recency) remains the default despite trailing the learned fusion by 16–17 pp in overall pairwise ranking accuracy on the held-out proxy; given the proxy’s sensitivity (Section 3.2), the fusion is user-selectable rather than default.



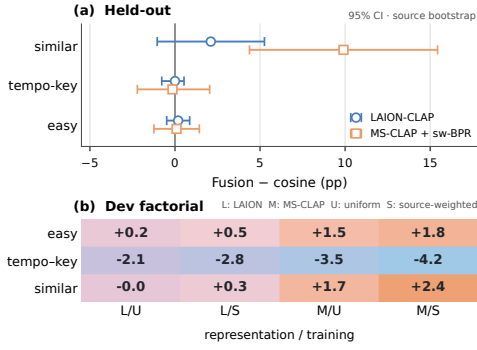


Figure 2. Configuration sensitivity of the fusion-cosine gap (pp). (a) Held-out five-seed means with source-level bootstrap 95% CIs (10,000 resamples). (b) Dev-set 2×2 representation/training factorial (L/M: LAION/MS-CLAP; U/S: uniform/source-weighted BPR).

2.2 Uncertainty-Aware Proposal Workflow

Ranking evidence stays inspectable (per-signal chips) and creative authority stays with the musician: candidates can be previewed, inserted via a single Nexus transaction, and undone, restoring the exact prior entity set (validated offline against the official Nexus document validator). Interaction events (preview, accept, reject, insert, undo) are logged with rank position to support future supervision.

2.3 Deployment Configuration

Deployment adapts the five-signal ranker to the musician’s library. It uses the MS-CLAP embedding space (licensing, CPU inference; offline evaluation uses LAION-CLAP Music) and *source-weighted BPR*: equal weight per source, hard_similar negatives $\times 2.0$ in the loss, seed-averaged weights, trained on all train sources. The key signal is zeroed online (Audiotool documents lack key metadata), and tag context comes from the session fragments rather than FSLD pack siblings.

3. COLD-START PROXY EVALUATION

3.1 Weak-Supervision Setup

With no interaction data at launch, the fusion trains on pack co-membership as a weak-supervision proxy. From the FSLD [9] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding uploader-only grouping; each pack is one *source*, the grouping unit for all splits. Each pack with ≥ 3 items yields one leave-one-out query per item: the left-out item is the positive, while 2–4 random pack siblings form the session stand-in. Cross-pack items provide negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest CLAP cosine to the positive from another pack). Two properties of this construction motivate Section 3.2: cross-pack negatives can be false negatives, most plausibly in hard_similar, and sibling tags correlate with pack membership, coupling the tag signal to the proxy label. We use a fixed 121-source held-out split (seed 20260810); LAION-CLAP Music [11] is the offline backbone. Development uses the remaining 484 train

Negative regime	LAION-CLAP	MS-CLAP, sw-BPR
easy	+0.2 [−0.5, +0.9]	+0.1 [−1.2, +1.4]
hard_tempo_key	+0.0 [−0.8, +0.5]	−0.1 [−2.2, +2.0]
hard_similar	+2.1 [−1.0, +5.3]	+9.9 [+4.4, +15.4]
overall	+0.6 [−0.4, +1.6]	+2.6 [+1.0, +4.2]

Table 1. Held-out fusion-cosine pairwise-accuracy delta (pp; 5-seed mean, source-level bootstrap 95% CI; sw-BPR: source-weighted). We report pairwise ranking accuracy: the fraction of positive-negative pairs where the positive scores higher (ties 0.5). Dev-side training uses 5 seeded source-grouped splits.

3.2 Three-Axis Sensitivity Analysis

The proxy captures curator-defined pack co-membership, not session compatibility; conclusions drawn from it shift along three axes. **Label construction.** An earlier corpus treated same-uploader items as co-members (26% of examples, 949 sources); fusion appeared +4.0 pp over frozen cosine on hard_similar. Restricting to true pack co-membership erases the dev-set hard_similar gain (+4.0 $\rightarrow$ −0.03 pp), and fusion trails cosine on hard_tempo_key by 2.1 pp (negative in all 5 splits); the earlier advantage may reflect a label-construction “horse” [12]. **Negative regime.** Leave-one-signal-out ablation on the dev set shows tempo helps on hard_similar (−0.96 pp when removed) but hurts on hard_tempo_key (+0.68 pp); key hurts both dev regimes: removing it adds +0.94/+1.21 pp (five-seed means, no CIs, descriptive). Held-out under MS-CLAP, tempo’s sign pattern recurs but key’s does not (−0.12 pp on hard_similar, sign unstable across seeds). The non-audio rules default falls to 0.424 pairwise accuracy on hard_tempo_key (0.787 overall, held-out, MS-CLAP column). **Representation/training configuration.** All held-out numbers here, including Table 1 and Figure 2, are exploratory and report corrected runs after a training-protocol mismatch was identified. Under LAION-CLAP the hard_similar advantage is +2.1 pp with a CI crossing zero; under MS-CLAP with source-weighted BPR (same source split; hard_similar re-mined in MS-CLAP space; hard_similar $\times 2.0$ in the loss) it is +9.9 pp. The two columns differ in embedding space, training recipe, and mined test negatives, so the 7.8 pp gap illustrates configuration sensitivity; a four-configuration dev-set factorial shows the same pattern (Figure 2b).

4. CONCLUSION

Pocket Producer frames DAW fragment retrieval as ranking for unobserved session compatibility under cold-start weak supervision. Conclusions about the fusion’s advantage over cosine change across label construction, negative regime, and representation/training configuration. Rather than treating proxy scores as compatibility judgments, the system surfaces retrieval as inspectable, reversible proposals. Future work will replace the cold-start proxy with interaction-derived evidence and evaluate retrieval against musicians’ creative choices.

5. ACKNOWLEDGEMENTS

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